INCH-POUND

MIL-P-48400A (AR)
AMENDMENT 5
23 August 1993
SUPERSEDING
AMENDMENT 4
03 October 1984

MILITARY SPECIFICATION

PROJECTILE, 60MM, HE, M720 (or M888),
METAL PARTS ASSEMBLY

This amendment forms a part of Military Specification MIL-P-48400A (AR) dated 11 July 1977, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "(or M888)" after M720; and wherever "M720" or "XM720" appears in specification.

2.1, Add the following new reference:

"FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings"

* Delete "MIL-I-6868" on this page and wherever it appears and substitute "MIL-STD-1949."

* Add "MIL-P-63194 Propellant, M10 for 60MM Mortar M204 Propelling Charge"

PAGE 2

2.1, Drawings: Delete "9275526 - Cartridge, 60MM, HE, M720".

Add: " 9236378 - Body Loading Assembly
11722522 - Pellet, Booster
11751196 - Fin M27".

Publications: Delete "P.A. Manual No. 5-1 - Fragmentation Testing Procedures".

AMSMC N/A FSC 1310

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

PAGE 3

* 3.2, Add to the end of the paragraph:

"Monitoring of the charts on each furnace used for heat treatment or stress-relief, as well as independent calibration of the furnaces, shall be performed at intervals approved by the Government, to ensure adequate control of temperature fluctuations and time at temperature."

* 3.2.1, Delete in its entirety.

* 3.3.1, Delete in its entirety.

* 3.4, Delete section in its entirety and substitute:

"3.4 Projectile defects.

3.4.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, solid shuts, pipes, porosity, inclusions, tears, blowholes, folds, seams, and other similar defects when inspected by magnetic particle inspection or "white" light in accordance with 4.5.2.1 and 4.5.2.2. Any projectile containing a discontinuity in which the metal at the bottom of the discontinuity cannot be seen throughout the zone shall be rejected.

PAGE 4

* 3.4.1.1 Body cavity. In addition to the requirements of 3.4.1, the body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, sharp edges and other similar defects.

* 3.4.2 Visual standards (internal and external). Visual standards for evaluation of metal defects after painting shall be established within 90 days after the start of production. However, the requirements for compliance with 3.4.1 and 3.4.1.1 are independent of visual standards."

* 3.6, Delete in its entirety and substitute:

"3.6 Proving ground tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

3.6.1 Maximum average pressure. There shall be no metal parts loss or break-up in the gun bore or in flight when tested in accordance with 4.5.4.1. Additionally, no round shall fall less than 80% of the average range of the other projectiles fired at maximum average pressure due to a projectile body flaw or unknown causes.

3.6.2 Service pressure. Projectile bodies shall reveal no evidence of failure when tested at service pressure in accordance with 4.5.4.2. Failure shall be:

a. Projectile body deformation in excess of 0.010 inch attributable to non-compliance with drawing or specification requirements. (For First Article only.)

b. Loss or breakup of any component, in the mortar tube or in flight.

c. A round falling less than 80% of the average range of the other projectiles fired at service pressure attributable to body non-compliance or unknown causes."

3.7.2, Delete paragraph in its entirety.

PAGE 5

3.9, Delete paragraph in its entirety.

* Add paragraph 3.11 as follows:

"3.11 Hydrotest. Each body shall successfully withstand an internal hydrostatic test load of 9500 + 500 PSI prior to machining the obturating band groove when tested in accordance with 4.5.7 without cracking, leakage, rupture or permanent deformation."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

* 4.1, Delete in its entirety and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

PAGE 6

* 4.3.1, Delete line (a) in its entirety and substitute "a. Maximum average pressure test ...4.5.4.1 ...5 complete projectile assemblies.

Delete line (b) in its entirety and substitute "b. service pressure test ...4.5.4.2 ...20 complete projectile assemblies.

4.3.3, Delete paragraph in its entirety and substitute:

"4.3.3 Rejection. See MIL-A-48078 (PA)."

* 4.4.1, Delete in its entirety and substitute:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following.

Source: https://assist.dla.mil -- Downloaded: 20260-22T19:24Z
Check the source to verify that this is the current version before use.

4.4.1.1 Stress-relieved lot. Stress-relieved lots shall contain bodies from not more than one mill heat of steel that have been manufactured and heat treated (including stress relief) to final mechanical properties, using the same equipment and the same temperature conditions in one unchanged process. No projectile shall be re-heat treated if it fails to meet the applicable mechanical properties requirements without Government approval. The lot thus constituted shall be the basis for the selection of samples for the hardness and tension tests herein after prescribed.

4.4.1.2 Projectile lots. Projectile lots shall consist of projectile assemblies produced by one manufacturer under one contract in one unchanged process, in accordance with the same drawing revision and specification revision, under conditions of a continuous production free of interruptions other than those due to the end of shift day or work week. All stress-relieved lots that have complied with the mechanical properties requirements of the applicable component drawing may be re-grouped for assembly into projectile lots. A projectile lot may contain projectiles from more than one mill heat of steel."

PAGE 7

* 4.4.2.1, Add to end of last sentence: "...and all material produced since the last successful inspection shall be rejected".

PAGE 8

* 4.4.2.2, Delete Critical 1 in its entirety and add "Critical 1 ...Hydrotest ...100% (AQL) ...3.11 1/ ...4.5.7."

* Add Special a. ...Metal defective ...2/ ...- ...3.4 ...Visual/4.5.2.1."

* Add "Note 1/ This test is performed prior to machining the obturator groove."

* Add "Note 2/ See paragraph 4.4.3.2."

PAGE 10

* 4.4.2.3, Critical 1, delete in its entirety.

* Add "Critical 2 ...Proving ground test ...1/ (No. of units) ...3.6 ...4.5.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

* Add "Special a. ...Metal defective ...- ...100% ...3.4 ...Visual/4.5.2.2..".

* Add "Note 1/ See 4.4.3.4."

PAGE 14

* 4.4.3.1.1, Delete section in its entirety and substitute:

"4.4.3.1.1 Hardness test. Projectiles bodies shall be hardness tested using the procedures specified in 4.5.1.1. Sampling procedures shall be as specified in 4.4.3.1.2."

* 4.4.3.1.2, Delete section in its entirety and substitute:

"4.4.3.1.2 Tension test. (See 4.5.1.3) Projectile bodies shall be hardness tested in accordance with 4.5.1.1 to confirm uniformity of the cold work/stress relief operations and to select the bodies for tensile testing. A minimum of twenty-five (25) bodies shall be selected from the first stress-relieved lot (See 4.4.1.1) from a non-continuous furnace, or from the first hour's production if a continuous furnace is used. These bodies shall be subjected to the hardness test of 4.5.1.1. The hardest and softest bodies from this group shall be sectioned and machined into tensile specimens for testing in accordance with 4.5.1.3.

If all tensile specimens from the hardest and softest bodies meet all mechanical property requirements, the hardness range represented shall be used as an accept/reject criteria for subsequent bodies processed from the same heat of steel and utilizing the same processing parameters as were used when processing the 25 selected bodies.

If any tensile specimen from the softest or hardest body fails to meet any mechanical property requirement, harder or softer bodies as required shall be selected and tensile tested as above until an acceptable hardness range is determined for that lot. The remainder of that stress-relieved lot shall be tested 100%; only bodies with hardness values which fall within values associated with acceptable mechanical properties (for the same mill heat and process parameters) shall be accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

If the hardest and softest of the 25 pass all mechanical property requirements, subsequent bodies from the stress-relieved lot shall be hardness tested at the rate of ten (10) per hour from a continuous furnace, or 20 per furnace load if a non-continuous heating unit is used. If all are within the acceptable hardness range of the initial 25 bodies, the production represented shall be considered acceptable. If a harder or softer body is found, that body shall be sectioned and tensile tested. If that body passes all mechanical property requirements, the acceptable hardness range will be extended accordingly; if it fails any requirement, the entire stress-relieved lot shall be screened and all bodies outside the acceptable limit shall be rejected.

If bodies processed from the same heat of steel utilize process parameters different from those of the original 25, those bodies shall constitute a new stress-relieved lot, and they shall be hardness and tensile tested as above in order to establish, widen, or limit the acceptable hardness range."

PAGE 15

* 4.4.3.1.3, Delete section in its entirety.

* 4.4.3.1.4.1, Sentence one: Delete "4.4.3.1.2.1" and substitute "4.4.3.1.2."

Change "stress relieved lot (or relief lot)" to "heat treatment batch" where it appears.

* 4.4.3.2, Delete in its entirety and substitute:

"4.4.3.2 Magnetic particle inspection. Testing shall be as specified in 4.5.2.1. After stress relief, hydrotesting and finish machining, bodies shall be inspected by magnetic particle inspection techniques approved by the responsible technical agency according to the following sequence:

a. 100 percent magnetic particle inspection will be in effect until 2500 consecutive bodies have been inspected and found to be free of metal defects.

b. Reduced inspection may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten consecutive working days.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

c. If no defect is found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive working days.

d. If no defect is encountered during the five days, the sample may be reduced to 5 bodies per hour.

e. If a defect is encountered by magnetic particle inspection during reduced sampling (b through d) or during final visual inspection after painting, the magnetic particle inspection shall revert back to (a) above. In addition a 100 percent back screening to the last acceptable hour's production will be performed at magnetic particle inspection.

f. For requalification of reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a through d) above.

g. Should a controversy arise as to the validity of an indication, the manufacturer may use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible technical agency.

h. Continual application of 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as an alternative procedure in lieu of corrective action to eliminate a continuing material or process induced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdraw acceptance inspection until the contractor can prove his process and materials are capable of producing suspect-free projectiles."

PAGE 16

* 4.4.3.3, Delete in its entirety and substitute:

"4.4.3.3 Salt spray test. (See 4.5.3) In performing the following tests, TT-C-490 shall apply unless specified otherwise herein. The tests shall be performed for first article inspection, each day of painting during production and whenever a change is made to the paint process, until all samples from five (5) consecutive days of painting have passed the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

Thereafter the sample frequency may be reduced to twice per week during continuous days of painting. Five specimens shall be selected, consisting of randomly chosen projectiles which have been processed through all the same cleaning, phosphating, painting and drying steps at the same time using the same equipment as other production projectile bodies. The samples shall be taken from each continuous final paint operation up to 24 hours long. One projectile shall be taken during the first thirty (30) minutes of painting, one during the last thirty (30) minutes and the others at any other time.

Prior to salt spray testing, two (2) of the specimens must only pass an adhesion test using the procedures and pass-fail criteria contained in 4.2.8 and 3.5.6 respectively, of TT-C-490. If both specimens pass the adhesion test, the three (3) remaining specimens shall be scored lengthwise using a precision scribing tool/insert or equivalent in preparation for the salt spray test."

* 4.4.3.4, Delete in its entirety and substitute:

"4.4.3.4 Proving ground tests. (See 3.6)

4.4.3.4.1 Maximum average pressure. (See 4.5.4.1) Five (5) projectiles shall be tested from the First Article sample, from each of the first three (3) production lots and from three (3) consecutive lots when a new mill heat of steel is introduced into production. Defect classification and individual accept/reject criteria shall be in accordance with Table II.

4.4.3.4.2 Service pressure. (See 4.5.4.2) Twenty (20) projectiles shall be tested from the First Article sample and ten (10) from each of the first three (3) production lots and from three (3) consecutive lots when a new mill heat of steel is introduced into production. For First Article only, an attempt will be made to recover at least 60 percent of the projectiles fired. Any projectile body which fails the requirement specified in 3.6.2a shall be forwarded to the technical agency (See 6.8). Defect classification and individual accept/reject criteria shall be in accordance with Table II.

Source: https://assist.dla.mil -- Downloaded: 20260-22T19:24Z
Check the source to verify that this is the current version before use.

Table II Proving Ground Test

Maximum Average Pressure Phase

Category	Defect	Accept	Reject
Critical	Premature burst or metal part breakup	0	1
Critical	Horizontal range less than 80% of average range	0	1

Service Pressure Phase

Category	Defect	Accept	Reject
Critical	Premature burst, metal part breakup (or deformation due to launch forces during First Article)	0	1
Critical	Horizontal range less than 80% of average range	0	1"

PAGE 17

4.5.3.5, Delete section in its entirety.

PAGE 18

4.4.3.6, Fourth sentence, change "The projectile assemblies shall be loaded and completely assembled with fuze in accordance with drawing 9275526 and MIL-C-48368" to read "The projectile bodies shall be loaded and assembled in accordance with 4.5.5.2."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

4.4.3.7, Delete in its entirety and substitute:

"4.4.3.7 Chemical composition. The chemical composition shall be as specified in ASTM-A322 as required on the applicable drawing when tested as specified in 4.5.6. The procedure for sampling shall be in accordance with ASTM-E350. Failure of any sample to meet the requirements is cause for rejection."

* 4.4.4, Add: "and 6.2 herein" to the end of the last sentence.

PAGE 19

* Delete paragraph 4.5.1.1 in its entirety and substitute:

"4.5.1.1 Hardness test. (See 4.4.3.1.1) The test shall be performed using any automated hardness tester which will produce accurate repeatable results with readings for an acceptable projectile between 25-80% of the full scale capability of the test equipment. When the Brinell hardness test is used, equipment and procedures shall be as prescribed in ASTM Method E-10, except that the minimum dwell time shall be 5 seconds, otherwise operating procedures shall be as specified by the applicable equipment manual, standard ASTM test methods and the following subject to approval by the Government representative:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a depth sufficient to remove the decarburization zone. (Note: Impressions shall be completely removed by machining of the obturating band groove or final machining.)

b. The hardness tester calibration shall be tested at start up, at least once each four hours of continuous operation or after a layoff of three or more hours.

c. The body shall be fixtured so that no rocking, shifting or deflection of the body will occur during the test.

d. The hardness tester shall not produce indentations deeper than 50% of the indentor ball diameter or 25% of the wall thickness of the projectile in that zone.

e. Hardness readings shall be taken at least ten indentor ball diameters apart. At least two readings shall be taken and averaged on each projectile."

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:24Z
Check the source to verify that this is the currentversion before use.

* 4.5.1.2, Delete in its entirety.

* 4.5.1.3, Delete in its entirety and substitute:

"4.5.1.3 Tension testing. From each sample selected for testing in accordance with 4.4.3.1.2, two tension test specimens (one bourrelet area, one boat tail area) conforming to the largest obtainable standard or subsize flat or round specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body, 180 degrees apart. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements. With the approval of the procuring agency, the Government shall not be restricted from testing additional specimens to determine that the mechanical properties meet the drawing requirements. Procedures and criteria for determining yield strength, elongation, reduction in area and tensile strength shall be as prescribed by ASTM Method E-8 except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0)."

* 4.5.1.4.1, Delete sentence two and substitute:

"The sample shall be removed from the hardest body(ies) undergoing tensile testing, from an area approximately 2.5 inches forward of the fin shoulder."

* Add new sentence at the end of the paragraph:

"If any specimen fails the 90 percent spherodization requirement, the applicable stress-relieved lot shall be rejected."

* 4.5.2, Delete paragraph in its entirety and substitute:

"4.5.2 Metal defects inspection.

4.5.2.1 Before paint (Magnetic particle inspection). Projectiles submitted in accordance with the sampling plan contained in 4.4.3.2 shall be inspected internally and externally for the defects described in 3.4. Projectiles shall be tested using procedures for the wet fluorescent method as specified in MIL-STD-1949. Equipment shall comply with MIL-M-6867. Each projectile shall be tested twice. Both transverse and

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

longitudinal fields shall be induced in the projectile using any method to establish the fields. Between the two (2) tests and following the second test, each projectile shall be demagnetized. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 mililiters (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 mils of solids per 100 mls suspension when water is used. The suspension shall be checked at the start of each days' production and at intervals of not over four hours of continuous operation, utilizing the procedures and equipment prescribed by section 4 of MIL-STD-1949. When performing the test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II. After draining the excess suspension, the body shall be inspected under ultraviolet light for the presence of defects. The requirement of MIL-STD-1949 to check system effectiveness by means of a steel ring specimen shall be waived. Components which exhibit indications which are suspected of being shallow surface defects may have material removed to a maximum of 0.005 inches subject to the following restrictions:

a. Involved areas shall be blended with adjacent surfaces to insure that no sharp edges, steps or gouges remain.

b. The involved areas shall be independently checked for verification that all drawing dimensions (wall thickness, variations, etc.) are within acceptable limits.

c. The components shall be again subjected to magnetic particle inspection for verification of defect removal.

* 4.5.2.2 <u>After paint ("white" light)</u>. Each projectile shall be visually inspected internally and externally for the defects described in 3.4 using any type of lighting which will facilitate good inspection of all surfaces. Standards shall be used when available. Detection of a body with a crack, solid shut, etc. at this inspection shall cause a return to 100% inspection at magnetic particle testing. Bodies exhibiting indications suspected of being shallow surface defects may be processed per 4.5.2.1, except that "white" light shall be used."

Source: https://assist.dla.mil -- Downloaded: 20260-22T19:24Z
Check the source to verify that this is the current version before use.

* 4.5.3, Delete in its entirety and substitute:

"4.5.3 Salt spray test. (See 4.4.3.3) The salt solution, test equipment and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise herein. Specimens shall be positioned between 15 and 30 degrees from vertical during the test. No coating of any kind shall be permitted over the paint on stamp marks of the specimens prior to testing, however areas in contact with racks or supports, unpainted surfaces and area within 3/16 inch of unpainted surfaces may be protected. When examining the specimens for signs of paint failure, the entire surface of the projectile shall be considered the significant surface.

As used herein, the term "blister" refers to an unbroken area of paint with a bubble-like appearance whereas, "pits" or "spots" are exposed phosphate pre-treatment or bare metal visible through holes in the paint. Blisters which are present when the test is over are to be counted within five (5) minutes of removal from the test chamber.

Projectiles shall be tested at the concentration and duration specified on Drawing 11751150. After testing, any projectile which fails for any of the following reasons shall cause rejection of all parts painted since the last successful test and a return to the original sampling plan if failure occurred during reduced sampling. The failure criteria in a-i applies to the significant surfaces only.

a. The paint shows more than 1/8 inch creepage, blistering or loss of adhesion of paint from the scribe mark, stamp mark, or edge.

b. There are more than five (5) blisters, total.

c. Any blister exceeds 1/16 inch average diameter.

d. Any pit exceeds 1/8 inch average diameter.

e. There are four (4) or more scattered pits exceeding 1/16 inch average diameter.

f. There are four (4) or more spots within one square inch.

g. There are three (3) spots within one square inch in two (2) or more places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

h. There are more than seven (7) scattered spots, total.

i. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 1/4 inch in diameter total area.

j. There is some variation of the above or an unlisted defect anywhere which the Government representative deems excessive."

* 4.5.4, Delete paragraph in its entirety and substitute:

"4.5.4 Proving ground tests.

4.5.4.1 Maximum average pressure. The metal parts assemblies provided for this phase shall be inert loaded to the service weight and assembled with dummy fuzes into complete rounds. Firing tests will be conducted in a mortar for which the round is standard, the tube of which will have a minimum of 25 percent life remaining based on current mortar tube condemnation criteria. The propelling charge will be adjusted to obtain 109 to 117 percent of the maximum average pressure of the propellant. The desired pressure (9363-10,050 PSI) must be reached with cartridges conditioned at 70 degrees F. See MIL-P-63194 for a table of propellant pressures. The test shall be performed at a Government proving ground in accordance with applicable U.S. Army TECOM acceptance test procedures. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6 and Table II in 4.4.3.4. The projectiles shall be recovered and examined to determine compliance with 3.6.

4.5.4.2 Service pressure. The metal parts assemblies provided shall be fired as in 4.5.4.1 except that the cartridges will be fired at charge 4; no adjustment of the propelling charge is required."

PAGE 21

4.5.5.1, Delete paragraph in its entirety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

4.5.5.2, Add "Load projectile in accordance with Drawing 9236378 and MIL-C-48368. Assemble a M27 Fin (Dwg 11751196) hand tight to aft end of loaded projectile. At forward end of projectile insert a booster pellet (Dwg 11722522) into fuze well cavity. Place a 1.25 inch OD x .24 inch ID x .75 thick cylindrical wooden plug over the booster pellet and insert a No. 6 blasting cap into the .24 inch diameter hole in the plug. Seat the blasting cap on the booster pellet and secure the assembly with tape making certain the pellet and blasting cap are fully seated."

4.5.6, Delete text and substitute: "The chemical composition shall be determined in accordance with the test procedures specified in ASTM-E350."

* Add new paragraph:

"4.5.7 Hydrotest. After the wall thickness is in conformance with final size but prior to machining the obturating band groove, each projectile shall be subjected to an internal hydrotest to determine compliance with the requirements of 3.11. Each projectile that successfully passes the hydrotest shall be stamped "H" 0.125 inch high by .01 inch deep approximately 0.35 inch below the location of the zone used for projectile nomenclature immediately after removal from the hydrotest equipment. Projectile bodies shall be tested using equipment capable of performing the test accurately and consistently. During the test the pressure will be applied gradually until the required minimum pressure is reached and held for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a lay-off of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the test, the bodies tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Projectile rupture, cracking, leakage or bulging during the test shall be cause for rejection of the body. The test fixture shall not restrict the projectile body from expanding longitudinally or radially under pressure. Protective barriers should be installed where applicable to shield operators and material from metal fragments which may come from defective projectiles that fail the test. If threads are engaged, the hydrotest plug should be designed with a one quarter inch minimum thread relief, to prevent loading of initial threads on the projectile body. Examination should be made of all hydrotest plugs after a projectile is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:24Z
Check the source to verify that this is the current version before use.

PAGE 22

* 6.2, Delete "DRDAR-QAR-I, Dover, New Jersey 07801" and substitute "Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-I, Bldg. 62S, Picatinny Arsenal, NJ 07806-5000.

This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply."

6.3, Delete paragraph in its entirety.

* 6.4, Add "The roughness comparison specimens prescribed by ANSI-B46.1 may be used as a basis for surface roughness determination when applicable."

* 6.7, Delete "ARRADCOM" and substitute "ARDEC". Delete address shown and substitute: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAN-Q, Bldg. 92, Picatinny Arsenal, NJ 07806-5000.

* 6.8, Delete in its entirety and substitute:

"6.8 Excess deformation. Forward suspect projectiles(s) to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-B, Bldg. 62S, Picatinny Arsenal, NJ 07806-5000."

PAGE 24

* Figure 1: Delete in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A538)

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:24Z
Check the source to verify that this is the currentversion before use.